ChangeMamba: Remote Sensing Change Detection With Spatiotemporal State Space Model

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***Abstract*— Convolutional neural networks (CNNs) and Transformers have made impressive progress in the field of remote sensing change detection (CD). However, both architectures have inherent shortcomings: CNN is constrained by a limited receptive field that may hinder their ability to capture broader spatial contexts, while Transformers are computationally intensive, making them costly to train and deploy on large datasets. Recently, the Mamba architecture, based on state space models (SSMs), has shown remarkable performance in a series of natural language processing tasks, which can effectively compensate for the shortcomings of the above two architectures. In this article, we explore for the first time the potential of the Mamba architecture for remote sensing CD tasks. We tailor the corresponding frameworks, called MambaBCD, MambaSCD, and MambaBDA, for binary CD (BCD), semantic CD (SCD), and building damage assessment (BDA), respectively. All three frameworks adopt the cutting-edge Visual Mamba architecture as the encoder, which allows full learning of global spatial contextual information from the input images. For the change decoder, which is available in all three architectures, we propose three spatiotemporal relationship modeling mechanisms, which can be naturally combined with the Mamba architecture and fully utilize its attribute to achieve spatiotemporal interaction of multitemporal features, thereby obtaining accurate change information. On five benchmark datasets, our proposed frameworks outperform current CNN- and Transformer-based approaches without using any complex training strategies or tricks, fully demonstrating the potential of the Mamba architecture in CD tasks. Further experiments show that our architecture is quite robust to degraded data. The source code is available at: https://github.com/ChenHongruixuan/MambaCD.**

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I. Introduction

CHANGE detection (CD) has been a popular field within the remote sensing community since the inception of remote sensing technology. It aims to detect changes in objects on the Earth's surface from multitemporal remote sensing images acquired at different times. Depending on the desired result from the change detector, CD tasks can be categorized into three types, namely, binary CD (BCD), semantic CD (SCD), and building damage assessment (BDA). Now, CD techniques play an important role in many fields, including land cover change analysis, urban sprawl studies, disaster response, geographic information system (GIS) updating, and ecological monitoring [1], [2], [3], [4], [5], [6], [7].

Optical high-resolution remote sensing imagery has become one of the most applied and researched data sources in the field of CD, which can provide detailed textural and geometric structural information about surface features, allowing us to achieve more refined changes [3]. Due to the increased heterogeneity within the same land-cover feature brought about by the increase in spatial resolution, traditional pixel-based CD methods [8], [9], [10] are difficult to achieve satisfactory detection results. To cope with this, researchers proposed object-based CD methods [11], [12], [13], which take the object consisting of homogeneous pixels as the basic unit for CD. However, these methods rely on hand-designed "shallow" features, which are not representative and robust enough to cope with complex ground conditions in multitemporal high-resolution images [14], [15].

The emergence of deep learning has brought new models and paradigms for CD, greatly improving the efficiency and accuracy of CD. After the early days of developing image patch-based approaches with tiny deep learning models [16], [17], [18], the release of a number of large-scale benchmark datasets [19], [20], [21] has allowed us to start designing and introducing larger and deeper models. Convolutional neural network (CNN)-based methods have been dominant within the field of CD since the introduction of fully convolutional networks (FCNs) to the field by Daudt et al. [22]. By employing models within the computer vision field as well as a priori knowledge of the CD task, many representative methods have

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been proposed [3], [4], [23], [24], [25], [26], [27]. Despite the decent results achieved by these methods, the inherent shortcomings of the CNN model structure, i.e., the inability of the limited receptive field to capture long-range dependencies between pixels, make these methods still fall short when dealing with complex and diverse multitemporal scenes in images with different spatial–temporal resolutions [28]. The emergence of the visual Transformer [29] provides an effective way to solve the above problems. By means of the stacked self-attention modules, the Transformer architecture can fully model the relationship between all the pixels of the whole image. Currently, more and more CD architectures adopt Transformer architectures as the encoder to extract representative and robust features [30], [31], [32], [33] and utilize it in the decoder to capture the spatiotemporal relationships between multitemporal features [28], [34], exhibiting superior performance.

Nonetheless, the self-attention operation demands quadratic complexity in terms of image sizes, resulting in expensive computational overhead, which is detrimental in dense prediction tasks, e.g., land-cover mapping, object detection, and CD, in large-scale remote sensing datasets. Some available solutions, such as limiting the size of the computational window or the step size to improve attention efficiency, come at the cost of imposing limits on the size of the reception field [35], [36]. As a viable alternative to the Transformer architecture, state space models (SSMs), especially structured state space sequence models (S4) [37], have shown cutting-edge performance in continuous long-sequence data analysis, with the promising attribute of linear scaling in sequence length. The Mamba architecture [38] further improves on the S4 model by employing a selection mechanism that allows the model to select relevant information in an input-dependent manner. By combining it with hardware-aware implementations, Mamba outperforms Transformers on a number of downstream tasks with high efficiency. Very recently, Mamba architecture has been extended to image data and shown promising results on some visual tasks [39], [40]. However, its potential for high-resolution optical images and different CD subtasks is still under-explored. How to design a suitable architecture based on a priori knowledge of CD tasks is undoubtedly valuable for subsequent research in the community.

In this article, we present the first attempt at introducing Mamba architecture for CD tasks and propose several spatiotemporal SSMs (collectively called ChangeMamba) to realize effective and robust CD. ChangeMamba can effectively model the global spatial context and spatiotemporal relationships in multitemporal images, thereby showing very promising results on the three subtasks of CD, namely, BCD, SCD, and BDA. Specifically, ChangeMamba is based on the recently proposed VMamba architecture [39], which adopts a cross-scan module (CSM) to unfold image patches in different spatial directions to achieve effective modeling of global contextual information from images. Since CD tasks require the detector to adequately learn spatiotemporal features from multitemporal images, we design three spatiotemporal relationship modeling mechanisms and naturally combine them with Mamba architecture. The proposed three mechanisms can adequately model the spatiotemporal relationships between multitemporal features by scanning them in different ways, and thus effectively detect different categories of changes, including binary changes, semantic changes, and building damage levels.

In summary, the main contributions of this article are as follows.

1) For the first time, we explore the application of the Mamba architecture in the field of remote sensing CD, thereby achieving high accuracy, efficient, and robust CD.
2) Based on the Mamba architecture and incorporating the unique characteristics of CD tasks, we design the corresponding network frameworks for each of the three CD tasks, i.e., BCD, SCD, and BDA.
3) Three spatiotemporal relationship modeling mechanisms are proposed, tailored for CD tasks, with the aid of the Mamba architecture to fully learn spatiotemporal features.
4) The proposed three frameworks show very competitive and even SOTA performance on five benchmark datasets, demonstrating their superiority. The source code for this work is publicly available for contributions to subsequent possible research.

The remainder of this article is organized as follows. Section II reviews related work. Section III details the three ChangeMamba architectures proposed for BCD, SCD, and BDA, respectively. The experimental results and discussion are provided in Section IV. Section V draws the conclusion.

II. Related Works

In this section, we focus on reviewing deep learning-based CD methods using optical high-resolution remote sensing images as data sources.

A. CNN-Based Method

The rapid evolution of CNN and its prowess in automatically extracting hierarchical features have led to wide applications in CD tasks. After the early stage of the patch-based framework [16], [18], Daudt et al. [22] presented the first attempt to introduce FCNs into the BCD field and three basic FCNs architectures were developed. After that, many representative network architectures were proposed. Zhang et al. [23] sought to tailor networks for semantic segmentation tasks more suitable for the BCD task by integrating feature change maps as a deep supervised signal, thus refining the final prediction. To make the framework suitable for handling multisource images, Chen et al. [17] proposed a framework by combining long- and short-term memory (LSTM) networks with CNNs. Furthermore, Fang et al. [41] designed a Siamese architecture with denser connections to facilitate more effective shallow information exchange for BCD. To combat the challenge of uneven foreground–background distributions within BCD, Han et al. [42] employed a novel sampling approach and an attention mechanism to integrate global information more effectively. After that, they introduced an innovative self-attention module that not only enhances

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long-distance dependency but also specifically targets the refinement of BCD predictions [43]. Zhao et al. [44] explored the effect of different fusion strategies on BCD performance and designed a dual encoder–decoder architecture to improve detection performance. To solve the problem of insufficient samples, the transfer learning technique was also introduced. Guo et al. [4] developed an attention-based network for semi-supervised building updating. Cao and Huang [25] designed a full-level fused cross-task transfer learning architecture for building CD. Recently, Chen et al. [45] have developed a sample generation method by simply exchanging image patches, which can train change detectors on single-temporal images in an unsupervised manner.

Compared with BCD, which only aims to identify "where" changes have occurred, SCD further seeks to determine "what" the changes are. Although this poses a greater challenge, it is more important for practical application analysis [24], [46]. Early explorations based on CNNs for SCD is the study [46], which adopts a multitask learning framework to simultaneously predict binary change maps and land cover maps, using the latter to assist in generating the final semantic change prediction. Mou et al. [47] proposed to combine recurrent neural network (RNN) and CNN for SCD. Yang et al. [20] introduced a benchmark dataset for SCD along with a new evaluation metric. Ding et al. [48] proposed a more powerful architecture to address the insufficient exchange of information between the semantic and change decoders. A new loss function was introduced to force the network to correlate two decoders. Zheng et al. [24] continued to strengthen the interactions between multitemporal features, designing a temporal-symmetric transformer (TST) module to model the spatiotemporal relationships. In [49], channel attention was used to embed change information into temporal features. In [50], temporal salient features were fused to extract and improve the representation of changes. In [14] and [51], unsupervised SCD methods were also studied, using pretrained CNNs or unsupervised learning techniques to extract hierarchical features, then comparing these features to determine changes.

Due to the lack of suitable benchmark datasets caused by the scarcity of disaster events and difficulty in annotation, the methodology progression of BDA has been relatively slow. The establishment of the xBD benchmark dataset in the xView2 Challenge [52], in turn, provides a large-scale benchmark for the field. The procedural approach to BDA typically encompasses two primary components: identifying the location of the building and assessing the level of damage incurred. In the xBD dataset, there are four damage levels, i.e., "no damage," "minor damage," "major damage," and "destroyed." This bifurcation essentially positions BDA as a nuanced subset of SCD, with a focus on quantifying damage levels. Among established approaches, the xView2 challenge [52] introduced a baseline method that employs a residual U-Net for building localization and damage classification, trained with pre- and post-disaster imagery. The Siamese U-Net, the winning approach of the xView2 challenge [52], utilized an U-Net framework for both pixel-level building localization and damage-level classification. It initialized the U-Net for the damage classification task with the parameters from the U-Net for the building localization task. A further advancement [53] utilized mask-RCNN architecture [54] to address BDA, incorporating Siamese encoders, multitask heads, and instance-level predictions. Zheng et al. [3] introduced a deep object-based architecture that encompasses shared and task-specific encoders, alongside an FCN-based multitask decoder. This development enhances task integration with an object-based postprocessing strategy, resulting in more refined outcomes.

The development of CNN-based CD methods has matured significantly over time. However, despite their excellent ability to extract local features, CNN architectures inherently struggle to capture long-distance dependencies [29]. This limitation becomes particularly critical in CD tasks, where changed regions, compared with background areas, are often sparse and scattered. The network's ability to model relationships beyond local areas is essential for accurately identifying these changed regions [28], [30].

B. Transformer-Based Method

Recently, promising results have been obtained by Transformers in the computer vision field [29], [36]. Their superior long-distance modeling capabilities have made them effectively overcome the challenges faced by the CNNs. As a result, many studies have begun to address CD tasks using Transformer architectures. In the field of BCD, Chen et al. [28] emerged as a pioneering approach, transforming multitemporal images into semantic tokens to model spatial–temporal relationships within a token-based framework, enhancing the CD process. Bandara and Patel [30] proposed a pure Transformer-based Siamese network, which has a Siamese Transformer encoder coupled with a multilayer perceptron (MLP) decoder. This architecture eliminates the need for a CNN-based feature extractor. Similarly, Zhang et al. [32] proposed a pure Transformer-driven model, adopting a Siamese U-shaped structure to fully recover detailed change information. Li et al. [34] introduced a hybrid model that blends the global contextual modeling capability of Transformers with the CNN block and U-Net architecture. It eliminates U-Net's redundant information and enhances feature quality through a difference enhancement module, leading to more precise detection results.

For Transformer-based SCD, Niu et al. [55] combined preactivated residual and Transformer blocks to extract semantic features from multitemporal image pairs, which is crucial for detecting subtle changes. The model's efficacy is further amplified by multitask prediction branches and a custom loss function. In addition, Ding et al. [56] introduced the semantic change Transformer, a CSWin Transformer adaptation, to explicitly learn semantic transitions. The model has a triple encoder–decoder architecture to enhance spatial features and incorporates spatiotemporal dynamic and task-specific prior information, leading to a decrease in learning disparities. This approach significantly boosts accuracy on benchmarks, adeptly managing scenarios with sparse labels.

Transformer-based BDA methods are currently understudied. Chen et al. [31] proposed the first pure Transformer-based

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BDA architecture called DamFormer. It has a Siamese Mix Transformer (MiT) [36] encoder and a lightweight all-MLP decoder to efficiently process multitemporal image pairs. DamFormer employs a multitemporal fusion module for improved information integration and a dual-task decoder for precise building localization and damage classification. By adopting strategies like weight sharing and innovative fusion mechanisms, it efficiently handles high-resolution imagery with lower computational costs, making it effective and streamlined for BDA.

Although Transformer-based methods have significantly enhanced CD accuracy through their powerful nonlocal modeling capabilities, they come with substantial computational resource demands. Specifically, as the number of tokens increases, the consumption of computational resources escalates quadratically, posing challenges for processing large-scale input, such as optical high-resolution imagery. Recently, the emerging Mamba architecture [38] not only rivals Transformers in nonlocal modeling capabilities but also maintains computational overhead at a linear growth rate, which opens up the possibility of efficient and accurate CD.

C. State Space Model

The SSM concept first popped up with the S4 model [37], offering a new way to handle contextual information globally, which caught attention due to its attractive property of scaling linearly in sequence length. Based on the S4 model, Smith et al. [57] proposed a new S5 model by introducing MIMO SSM and efficient parallel scan into the S4 model. After that, the H3 model [58] further advanced these concepts, achieving a performance on par with that of Transformers in language modeling tasks. Recently, Gu and Dao [38] proposed a data-dependent SSM layer and built a generic language model backbone called Mamba, which outperforms Transformers at different scales on large-scale real data and scales linearly in sequence length. Very recently, Visual Mamba [39] and Vision Mamba [40] have extended the Mamba architecture to 2-D image data, showing superior performance on many computer vision tasks. Inspired by this progress, some pioneering work has introduced the Mamba architecture to the field of remote sensing image processing [59], [60]. Their studies show that the Mamba architecture can yield performance comparable to the advanced CNN and Transformer architectures on scene classification [60] and pansharpening [59] tasks. However, these works still focus on low-level tasks or classification tasks, and all of these tasks are single-temporal tasks [45]. The potential of the Mamba architecture for multitemporal remote sensing image-related scenarios and dense prediction tasks remains to be explored.

III. Methodology

A. Preliminaries

The SSM-based models and Mamba are inspired by linear time-invariant systems, which map a 1-D function or sequence $x(t) \in \mathcal{R}$ to the response $y(t) \in \mathcal{R}$ through a hidden state $h(t) \in \mathcal{R}^N$. These systems are usually formulated as linear ordinary differential equations (ODEs) as follows:

$$\begin{cases} h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) = \mathbf{C}h(t) \end{cases} \tag{1}$$

where $\mathbf{A} \in \mathcal{R}^{N\times N}$ is the evolution parameter and $\mathbf{B} \in \mathcal{R}^{N\times 1}$, $\mathbf{C} \in \mathcal{R}^{1\times N}$ are the projection parameters.

The S4 model [37] is a discrete counterpart of the continuous system, as continuous systems face significant challenges when integrated into deep learning algorithms. It contains a time scale parameter Δ that is used to convert the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ into their discrete counterparts $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$. A prevalent way to achieve this transformation is to use a zero-order hold (ZOH) approach

$$\begin{cases} \overline{\mathbf{A}} = \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I})\Delta\mathbf{B}. \end{cases} \tag{2}$$

After that, the discretization of (1) can be formulated as follows:

$$\begin{cases} h'_t = \overline{\mathbf{A}}h_{t-1} + \quad \overline{\mathbf{B}}x_t \\ y_t = \mathbf{C}h_t. \end{cases} \tag{3}$$

The output of x with sequence length $\mathbf{L}_x$ can be calculated directly by the following global convolution operation:

$$\begin{cases} \overline{\mathbf{K}} = \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{\mathbf{L}_x-1}\overline{\mathbf{B}}\right) \\ \mathbf{y} = \mathbf{x} * \overline{\mathbf{K}} \end{cases} \tag{4}$$

where $\overline{\mathbf{K}} \in \mathcal{R}^{\mathbf{L}_x}$ is a structured convolutional kernel and $*$ denotes a convolutional operation.

Mamba architecture [38] builds on the S4 model by proposing a selection mechanism that allows the model to filter out irrelevant information and recall relevant information.[1] In addition, a hardware-aware algorithm has been developed for the recursive computation of the model but will not be implemented in the extended state, optimizing the GPU memory layout. These characteristics allow the Mamba architecture not only to obtain promising results on languages and other data with long sequences but also to achieve fast training and inference with memory and computational overhead linearly proportional to the sequence length.

B. Problem Statement

In this article, we focus on three subtasks within the CD field. They are BCD, SCD, and BDA. The definitions of the three tasks are as follows.

1) Binary Change Detection: BCD is the basic and the most intensively studied task in the CD field. It focuses on "where" change occurs. According to the category of interest, BCD can also be divided into category-agnostic CD focusing on general land-cover change information, and single-category CD, e.g., building CD and forest CD. Given a training set represented as $\mathcal{D}^{\text{bcd}}_{\text{train}} = \{(\mathbf{X}^{T_1}_i, \mathbf{X}^{T_2}_i, \mathbf{Y}^{\text{bcd}}_i)\}^{N^{\text{bcd}}_{\text{train}}}_{i=1}$, where $\mathbf{X}^{T_1}_i, \mathbf{X}^{T_2}_i \in \mathcal{R}^{H\times W\times C}$ is the ith multitemporal image pair acquired in T_1 and T_2, respectively, and $\mathbf{Y}^{\text{bcd}}_i \in \{0, 1\}^{H\times W}$ is the corresponding

[1] Mamba architectures sometimes are abbreviated as the S6 model since they are S4 models with a selection mechanism and computed with a scan [38].

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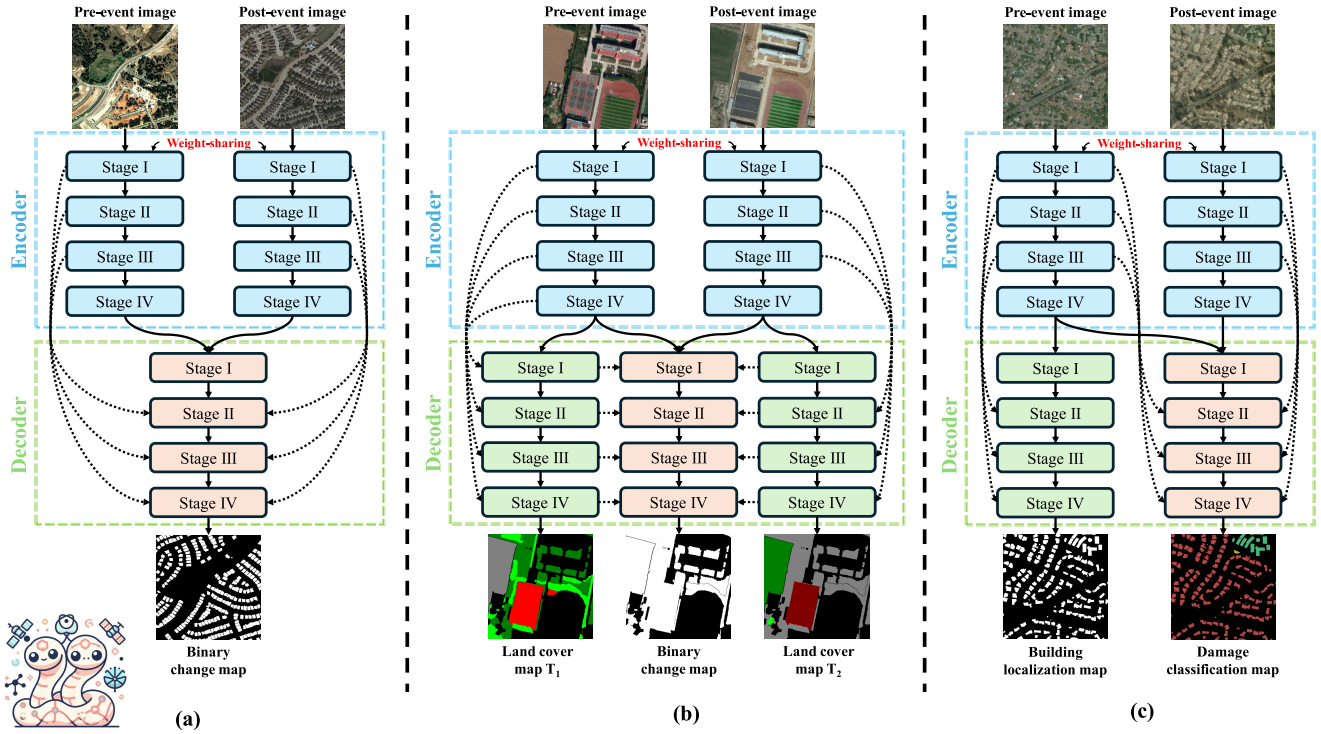


Fig. 1. Network framework of the proposed (a) MambaBCD for BCD, (b) MambaSCD for SCD, and (c) MamabaBDA for BDA. The specific structure of the encoder and decoder is shown in Figs. 2, 4, and 6.

label, the goal of BCD is to train a change detector $\mathcal{F}_\theta^{\text{bcd}}$ on $\mathcal{D}_{\text{train}}^{\text{bcd}}$ that can predict change maps reflecting accurate "change/nonchange" binary information on new sets.

2) Semantic Change Detection: Compared with BCD, SCD focuses not only on "where" the change occurs but also on the "what" of the change, i.e., "from–to" semantic change information [48]. The training set in the SCD task can be represented as $\mathcal{D}_{\text{train}}^{\text{scd}} = \{(\mathbf{X}_i^{T_1}, \mathbf{X}_i^{T_2}, \mathbf{Y}_i^{T_1}, \mathbf{Y}_i^{T_2}, \mathbf{Y}_i^{\text{bcd}})\}_{i=1}^{N_{\text{train}}}$. Compared with BCD, the corresponding land-cover labels $\mathbf{Y}_i^{T_1}, \mathbf{Y}_i^{T_2} \in \{0, 1, \ldots, \mathbf{C}^{\text{lcm}}\}^{H\times W}$ with $\mathbf{C}^{\text{lcm}}$ categories of $\mathbf{X}_i^{T_1}$ and $\mathbf{X}_i^{T_2}$ are additionally required. The goal of SCD is to train a semantic change detector $\mathcal{F}_\theta^{\text{scd}}$ on $\mathcal{D}_{\text{train}}^{\text{scd}}$ that can predict land-cover maps of multitemporal images and binary change maps between them on new sets as accurately as possible. By combining the information in predicted land-cover maps and binary change maps, the "from–to" semantic change information can be derived.

3) Building Damage Assessment: BDA is a special "one-to-many" SCD task [3]. BDA requires recognizing not only the "where" of the change (damage) occurs but also the postevent states of the object of interest (building) given the pre-event state of the object of interest. The training set in the BDA task can be denoted by $\mathcal{D}_{\text{train}}^{\text{bda}} = \{(\mathbf{X}_i^{T_1}, \mathbf{X}_i^{T_2}, \mathbf{Y}_i^{\text{loc}}, \mathbf{Y}_i^{\text{clf}})\}_{i=1}^{N_{\text{train}}}$, where $\mathbf{Y}_i^{\text{loc}} \in \{0, 1\}^{H\times W}$ is the label of the object-of-interest (building) at T_1, and $\mathbf{Y}_i^{\text{clf}} \in \{0, 1, \ldots, \mathbf{C}^{\text{dam}}\}^{H\times W}$ is the postevent state (damage level) of the object of interest at T_2.

C. Network Architecture

According to the commonalities and properties of the three CD subtasks, we design the corresponding network frameworks based on the Mamba architecture, called MambaBCD, MambaSCD, and MamabBDA, respectively. Fig. 1 shows the network framework of these three architectures. Among them, the encoders of the three networks are weight-sharing siamese networks based on the Visual State Space Model (VMamba) architecture [39]. VMamba can adequately extract the robust and representative features of the input images that benefit from the Mamba architecture and an efficient 2-D cross-scan mechanism (as shown in Fig. 3) for downstream tasks. All three architectures have a change detector of the same structure for learning spatiotemporal relationships from the features extracted by the encoder. In addition, MambaSCD and MambaBDA additionally possess semantic decoders for other tasks, i.e., land-cover classification and building localization. The specific structure of the encoders and decoders in our architectures will be elaborated in Sections III-D and III-E. In this section, we focus on the input, output, and internal information flow of these three network architectures.

1) MambaBCD: MambaBCD is the architecture designed for the BCD task. First, the Siamese encoder network denoted by $\mathcal{F}_\theta^{\text{enc}}$ extracts multilevel features from the input multitemporal images as $\{\mathbf{F}_{i,j}^{T_1}\}_{j=1}^4 = \mathcal{F}_\theta^{\text{enc}}(\mathbf{X}_i^{T_1})$ and $\{\mathbf{F}_{i,j}^{T_2}\}_{j=1}^4 = \mathcal{F}_\theta^{\text{enc}}(\mathbf{X}_i^{T_2})$. Next, these multilevel features are fed into a tailored change decoder $\mathcal{F}_\theta^{\text{cdec}}$. Based on the Mamba architecture, the change decoder can fully learn the spatiotemporal relationship from multilevel features through three different mechanisms, and gradually obtains an accurate BCD result, formulated as $\mathbf{P}_i^{\text{bcd}} = \mathcal{F}_\theta^{\text{cdec}}(\{\mathbf{F}_{i,j}^{T_1}\}_{j=1}^4, \{\mathbf{F}_{i,j}^{T_2}\}_{j=1}^4)$. The binary change map can be obtained as $\hat{\mathbf{Y}}_i^{\text{bcd}} = \text{argmax}_c \mathbf{P}_i^{\text{bcd}}$.

2) MambaSCD: MambaSCD is the architecture designed for the SCD task. As shown in Fig. 1(b), MambaSCD adds two additional semantic decoders for land cover mapping tasks

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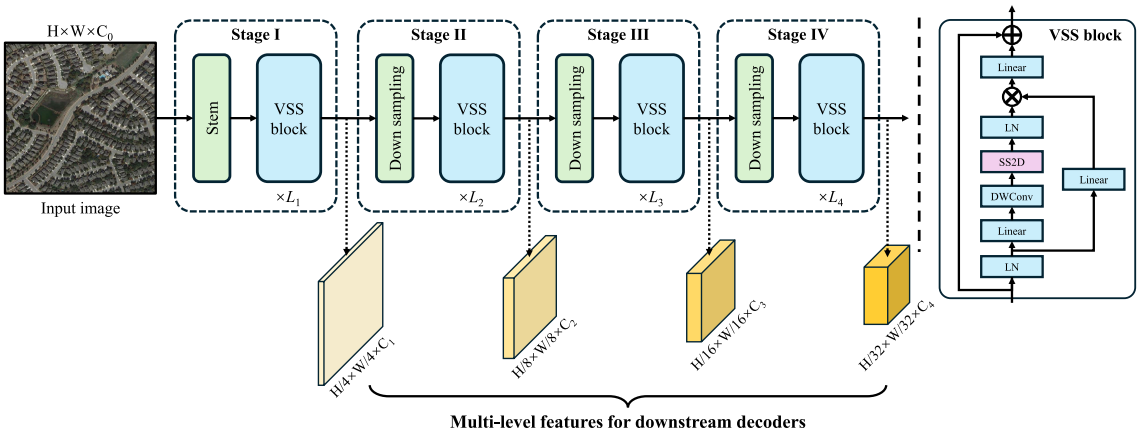


Fig. 2. Specific structure of the encoder network is based on the visual SSM used in our three architectures.

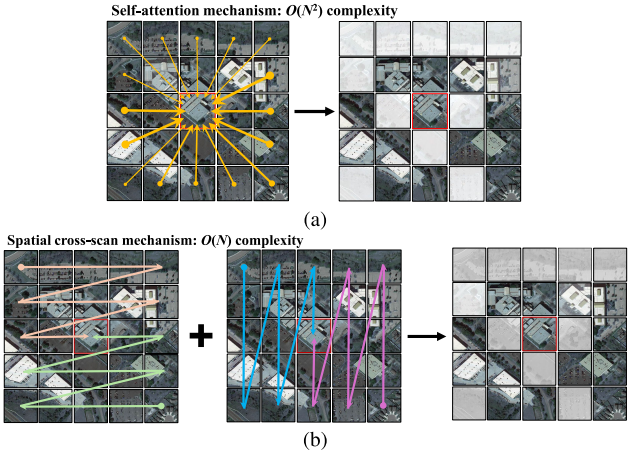


Fig. 3. Illustration of the (a) self-attention mechanism [29], [61] and (b) cross-scan mechanism [39] to capture global contextual information.

based on MambaBCD, denoted by $\mathcal{F}_\theta^{T_1}$ and $\mathcal{F}_\theta^{T_2}$. In addition to being treated as input as $\mathcal{F}^{\text{cdec}}$ to learn spatiotemporal relationships to predict BCD results, the multilevel features extracted by the encoder are also fed into $\mathcal{F}_\theta^{T_1}$ and $\mathcal{F}_\theta^{T_2}$ to predict the land-cover map of the corresponding temporal image, formulated as $\mathbf{P}_i^{T_1} = \mathcal{F}_\theta^{T_1}(\{\mathbf{F}_{i,j}^{T_1}\}_{j=1}^4)$ and $\mathbf{P}_i^{T_2} = \mathcal{F}_\theta^{T_2}(\{\mathbf{F}_{i,j}^{T_2}\}_{j=1}^4)$. After obtaining the land-cover maps $\hat{\mathbf{Y}}_i^{T_1} = \text{argmax}_c \mathbf{P}_i^{T_1}$ and $\hat{\mathbf{Y}}_i^{T_2} = \text{argmax}_c \mathbf{P}_i^{T_2}$ and the binary change map $\hat{\mathbf{Y}}_i^{\text{bcd}}$, the semantic change information of $T_1 \rightarrow T_2$ can be obtained by performing the mask operation on $\hat{\mathbf{Y}}_i^{T_1}$ and $\hat{\mathbf{Y}}_i^{T_2}$ using $\hat{\mathbf{Y}}_i^{\text{bcd}}$ [6], [24], [46].

3) MambaBDA: MambaBDA is the architecture presented for the BDA task. Unlike the typical SCD task, the BDA task only needs to predict land-cover maps (building location maps) for pre-event images. Therefore, in the MambaBDA architecture shown in Fig. 1(c), there is only one semantic decoder $\mathcal{F}_\theta^{\text{loc}}$ to predict the building localization map as $\mathbf{P}_i^{\text{loc}} = \mathcal{F}_\theta^{\text{loc}}(\{\mathbf{F}_{i,j}^{T_1}\}_{j=1}^4)$. Like MambaBCD and MambaSCD, a change decoder denoted by $\mathcal{F}_\theta^{\text{clf}}$ then learns spatiotemporal relationships from multitemporal features to classify the damage level of buildings as $\mathbf{P}_i^{\text{clf}} = \mathcal{F}_\theta^{\text{clf}}(\{\mathbf{F}_{i,j}^{T_1}\}_{j=1}^4, \{\mathbf{F}_{i,j}^{T_2}\}_{j=1}^4)$. The obtained building location map $\hat{\mathbf{Y}}_i^{\text{loc}} = \text{argmax}_c \mathbf{P}_i^{\text{loc}}$ can be used for further postprocessing on the obtained damage classification map $\hat{\mathbf{Y}}_i^{\text{clf}} = \text{argmax}_c \mathbf{P}_i^{\text{clf}}$ to improve the accuracy, e.g., object-based postprocessing methods [3].

D. Encoder Based on Visual State Space Model

Mamba architecture causally processes the input data and thus can only capture information within the scanned part of the data. This attribute naturally aligns with natural language data, which poses significant challenges when adapting to noncausal data, such as image [39]. Directly expanding image data along a certain dimension of space and inputting it into the S6 model will result in a lack of adequately modeled spatial contextual information. Recently, VMamba solved this problem well by proposing a 2-D cross-scan mechanism. As shown in Fig. 3, before inputting the tokens into the S6 model, the cross-scan mechanism rearranges the tokens in spatial dimensions, i.e., top-left to bottom-right, bottom-right to top-left, top-right to bottom-left, and bottom-left to top-right. Finally, the resulting features are then merged. In this way, any pixel can get spatial context information from different directions. Moreover, the computational complexity of the VMamba model under the cross-scan mechanism is still $O(N)$ compared with the self-attention operation in the Transformer.

Based on the VMamba architecture, the specific structure of the encoder network in our three proposed architectures is shown in Fig. 2. There are four stages, each of which first downsamples the input data, then fully models the spatial contextual information using a number of visual state space (VSS) blocks, and then outputs the features for that stage, i.e., $\mathbf{F}_{i,j}^{T_1}$ and $\mathbf{F}_{i,j}^{T_2}$. The structure of the VSS block is also depicted in Fig. 2. The input first passes through a linear embedding layer, and the output is split into two information flows. One flow passes through a 3×3 depthwise convolution (DWConv) layer, followed by a Silu activation function [62] before entering the core SS2D module (i.e., the integration of the S6 model with the cross-scan mechanism). The output of the SS2D module passes through a layer normalization (LN) layer and is then summed up with the outputs of the other streams that have been activated by Silu. This combination produces the final output of the VSS block. Finally, the features extracted for pre-event and postevent images from

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the four stages $\{\mathbf{F}_{i,j}^{T_1}\}_{j=1}^4$ and $\{\mathbf{F}_{i,j}^{T_2}\}_{j=1}^4$ are then used in the subsequent decoders responsible for specific tasks.

E. Task-Specific Decoders

1) Spatiotemporal Relationship Modeling Mechanism: Although the employed encoder can extract robust and representative features, learning spatiotemporal relationships of multitemporal images is also significant for CD tasks [24], [47]. To this end, we propose three mechanisms for modeling spatiotemporal relationships that can be aligned with the attribute of the S6 model capable of adequately modeling the global contextual information of long sequential data. Fig. 5 illustrates the three modeling mechanisms. They are spatiotemporal sequential modeling, spatiotemporal cross modeling, and spatial–temporal parallel modeling. Sequential modeling is unfolding the tokens of two temporal phases of data and then arranging them in temporal order as $\mathbf{F}_{i,j}^{\text{seq}} = [\mathbf{F}_{i,j}^{T_1}(1), \ldots, \mathbf{F}_{i,j}^{T_1}(HW/(2^{1+j})), \mathbf{F}_{i,j}^{T_2}(1), \ldots, \mathbf{F}_{i,j}^{T_2}(HW/(2^{1+j}))]$. The cross-modeling mechanism, on the other hand, is modeling by cross-ordering the tokens of the two temporal phases as $\mathbf{F}_{i,j}^{\text{crs}} = [\mathbf{F}_{i,j}^{T_1}(1), \mathbf{F}_{i,j}^{T_2}(1), \ldots, \mathbf{F}_{i,j}^{T_1}(HW/(2^{1+j})), \mathbf{F}_{i,j}^{T_2}(HW/(2^{1+j}))]$. Finally, parallel modeling is done by concatenating the tokens of the two temporal phases in the channel dimension and then performing joint modeling, i.e., $\mathbf{F}_{i,j}^{\text{pra}} = \text{concat}_c(\mathbf{F}_{i,j}^{T_1}, \mathbf{F}_{i,j}^{T_2})$. Through these three mechanisms and Mamba architecture, the spatiotemporal relationships intrinsic in the multitemporal features will be fully explored, helping the change decoder to obtain accurate CD results.

2) Change Decoder: Based on the proposed three spatiotemporal learning mechanisms, the specific structure of the change decoder is then shown in Fig. 4. It fully learns the spatiotemporal relationships from the extracted multitemporal features $\{\mathbf{F}_{i,j}^{T_1}\}_{j=1}^4$ and $\{\mathbf{F}_{i,j}^{T_2}\}_{j=1}^4$ in the four stages and obtains accurate binary change maps. At the beginning of each stage, the spatiotemporal relationship of multitemporal features is first modeled using a spatiotemporal state space (STSS) block. In the STSS block, a spatiotemporal tokens generator module will rearrange the input multitemporal features, which are then fed into three VSS blocks. Each block is responsible for learning one of the spatiotemporal relationships in Fig. 5. The output of the STSS block of the current stage is then integrated with the information from the feature map of the previous stage through a fusion module. In the fusion module, the number of channels of the low-level feature map is mapped to be consistent with the high-level feature map by a 1×1 convolutional layer. Then, the high-level and low-level feature maps are summed. Finally, the resulting feature map is smoothed by a residual layer. After passing through an upsampling layer, the feature map is fed to the next stage.

3) Semantic Decoder: The specific structure of the semantic decoder is shown in Fig. 6. It is mainly responsible for gradually recovering the land-cover maps from the corresponding multilevel features extracted by the encoder. It also has four stages. At the beginning of each stage, the global spatial context information of the input data are the first modeled using a VSS block. Then, the feature map is upsampled and integrated with information about the lower level feature map with higher resolution through a fusion module. In the last stage, the features output by the VSS block is upsampled four times and used to predict the corresponding land cover map.

F. Loss Function

Since this article focuses on exploring the potential of the Mamba architecture for CD tasks, we have optimized networks using commonly used loss functions in the field. Some tricks for improving accuracy such as focal loss [63], deep supervision [23], and multiscale training/testing [6] are not adopted. Even so, we can find that the proposed architecture achieves very competitive and even SOTA performance on all three subtasks in the experimental section.

1) BCD: Since BCD can be regarded as a special semantic segmentation task [45], we directly optimize the network using the cross-entropy loss as follows:

$$\mathcal{L}_{\text{ce}}^{\text{bcd}} = -\frac{1}{N_{\text{train}}^{\text{bcd}}} \sum_{i=1}^{N_{\text{train}}^{\text{bcd}}} \sum_{c=0}^{1} \tilde{\mathbf{Y}}_i^{\text{bcd}}(c) \log\left(\mathbf{P}_i^{\text{bcd}}(c)\right) \tag{5}$$

where $\tilde{\mathbf{Y}}_i^{\text{bcd}}$ is the one-hot form of $\mathbf{Y}_i^{\text{bcd}}$, $\mathbf{P}_i^{\text{bcd}} = \mathcal{F}_\theta^{\text{bcd}}(\mathbf{X}_i^{T_1}, \mathbf{X}_i^{T_2})$ is the output of the binary change detector through the softmax activation.

In addition, the Lovasz-softmax loss [64] is introduced to alleviate the problem of imbalance in the number of samples between changed and unchanged pixels. The final loss function for the BCD task is formulated as follows:

$$\mathcal{L}_{\text{final}}^{\text{bcd}} = \mathcal{L}_{\text{ce}}^{\text{bcd}} + \mathcal{L}_{\text{lov}}^{\text{bcd}}. \tag{6}$$

2) SCD: For the SCD task, in addition to optimizing the BCD task, it is also necessary to optimize the land cover mapping task for both pre-event and postevent images, which can be optimized with cross-entropy loss as follows:

$$\begin{cases} \mathcal{L}_{\text{ce}}^{T_1} = -\dfrac{1}{N_{\text{train}}^{\text{scd}}} \displaystyle\sum_{i=1}^{N_{\text{train}}^{\text{scd}}} \sum_{c=0}^{\mathbf{C}^{\text{lcm}}} \tilde{\mathbf{Y}}_i^{T_1}(c) \log\left(\mathbf{P}_i^{T_1}(c)\right) \\ \mathcal{L}_{\text{ce}}^{T_2} = -\dfrac{1}{N_{\text{train}}^{\text{scd}}} \displaystyle\sum_{i=1}^{N_{\text{train}}^{\text{scd}}} \sum_{c=0}^{\mathbf{C}^{\text{lcm}}} \tilde{\mathbf{Y}}_i^{T_2}(c) \log\left(\mathbf{P}_i^{T_2}(c)\right) \end{cases} \tag{7}$$

where $\mathbf{P}_i^{T_1}$ and $\mathbf{P}_i^{T_2}$ are the probabilistic map of land-cover classification for the pre-event image and postevent image output by the semantic decoder of the MambaSCD architecture and $\mathcal{L}_{\text{ce}}^{T_1}$ and $\mathcal{L}_{\text{ce}}^{T_2}$ are the corresponding cross-entropy loss.

Similarly, the Lovasz-softmax loss function is used to solve the sample imbalance problem. Thus, the final loss function for the SCD task takes the form

$$\mathcal{L}_{\text{final}}^{\text{scd}} = \mathcal{L}_{\text{ce}}^{\text{bcd}} + \mathcal{L}_{\text{lov}}^{\text{bcd}} + \frac{1}{2}\left(\mathcal{L}_{\text{ce}}^{T_1} + \mathcal{L}_{\text{ce}}^{T_2} + \mathcal{L}_{\text{lov}}^{T_1} + \mathcal{L}_{\text{lov}}^{T_2}\right). \tag{8}$$

3) BDA: BDA contains a building location task and a damage classification task. The cross-entropy loss for each task can be expressed as follows:

$$\begin{cases} \mathcal{L}_{\text{ce}}^{\text{loc}} = -\dfrac{1}{N_{\text{train}}^{\text{bda}}} \displaystyle\sum_{i=1}^{N_{\text{train}}^{\text{bda}}} \sum_{c=0}^{1} \tilde{\mathbf{Y}}_i^{\text{loc}}(c) \log\left(\mathbf{P}_i^{\text{loc}}(c)\right) \\ \mathcal{L}_{\text{ce}}^{\text{clf}} = -\dfrac{1}{N_{\text{train}}^{\text{bda}}} \displaystyle\sum_{i=1}^{N_{\text{train}}^{\text{bda}}} \sum_{c=0}^{\mathbf{C}^{\text{dam}}} \tilde{\mathbf{Y}}_i^{\text{clf}}(c) \log\left(\mathbf{P}_i^{\text{clf}}(c)\right) \end{cases} \tag{9}$$

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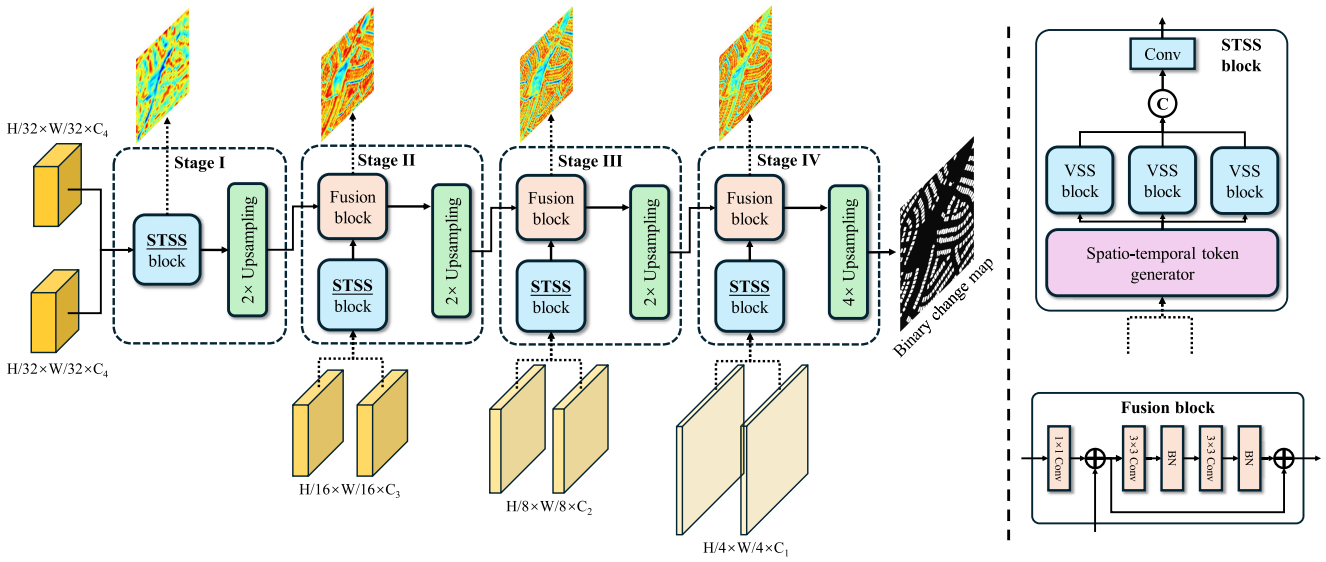


Fig. 4. Specific structure of the change decoder used in MambaBCD, MambaSCD, and MambaBDA architectures based on the proposed spatiotemporal relationship modeling mechanisms in Fig. 5.

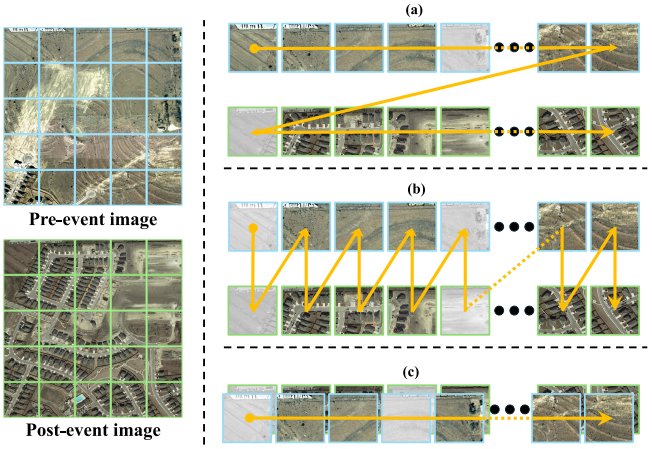


Fig. 5. Illustration of three mechanisms for learning spatiotemporal relationships. (a) Spatiotemporal sequential modelling. (b) Spatiotemporal cross modelling. (c) Spatiotemporal parallel modelling.

where $\mathbf{P}_i^{\text{loc}}$ and $\mathbf{P}_i^{\text{clf}}$ are the output of localization branch and classification branch of the MambaBDA architecture, respectively, and $\mathcal{L}_{\text{ce}}^{\text{loc}}$ and $\mathcal{L}_{\text{ce}}^{\text{clf}}$ are the corresponding cross-entropy loss of building localization and damage classification tasks, respectively.

By utilizing the Lovasz-softmax loss function to solve the sample imbalance problem, the final loss function of BDA is formulated as follows:

$$\mathcal{L}_{\text{final}}^{\text{bda}} = \mathcal{L}_{\text{ce}}^{\text{loc}} + \mathcal{L}_{\text{lov}}^{\text{loc}} + \mathcal{L}_{\text{ce}}^{\text{clf}} + \mathcal{L}_{\text{lov}}^{\text{clf}}. \tag{10}$$

IV. Experiments and Discussion

A. Datasets

1) SYSU-CD [19]: This dataset is a category-agnostic CD dataset, which introduces a comprehensive collection of 20 000 pairs of 0.5 m/pixel resolution aerial images from Hong Kong, spanning 2007–2014. This dataset is distinguished by its focus on urban and coastal changes, featuring high-rise buildings and infrastructure developments. For an effective deep learning application, the dataset is structured into training, validation, and test sets following a 6:2:2 ratio, with 20 000 patches of 256 × 256. This dataset encompasses a range of various change scenarios, such as urban construction, suburban expansion, groundwork, vegetation changes, road expansion, and sea construction.

2) LEVIR-CD + [65]: This advanced version of the LEVIR-CD dataset is a building CD dataset. It comprises 985 pairs of very high-resolution images at 0.5 m/pixel, each with dimensions of 1024 × 1024 pixels. Spanning a time interval of 5–14 years, these multitemporal images document significant building construction changes. It also encompasses a wide array of building types, including villa residences, tall apartments, small garages, and large warehouses, with a focus on both the emergence of new buildings and the decline of existing structures.

3) WHU-CD [66]: The WHU-CD dataset, a subset of the larger WHU Building dataset, is tailored for the building CD task. It comprises two aerial datasets from Christchurch, New Zealand, captured in April 2012 and 2016, with a spatial resolution of 0.3 m/pixel. This dataset is particularly focused on detecting changes in large and sparse building structures. The aerial images captured in 2012 cover an area of 20.5 km^2, featuring 12 796 buildings, while the image captured in 2016 shows an increase to 16 077 buildings within the same area, reflecting significant urban development over the four-year period. We follow the official protocol [66] to split the dataset into training and testing areas for experiments.

4) SECOND [20]: The SECOND dataset introduces a collection of 4662 pairs of 512 × 512 aerial images at 0.5–3 m/pixel resolution, annotated for SCD in cities such as Hangzhou, Chengdu, and Shanghai. It focuses on six primary land cover classes: nonvegetated ground surfaces, trees, low vegetation, water, buildings, and playgrounds to capture a wide range of geographical changes. The dataset uses land-cover map pairs and nonchange masks to accurately

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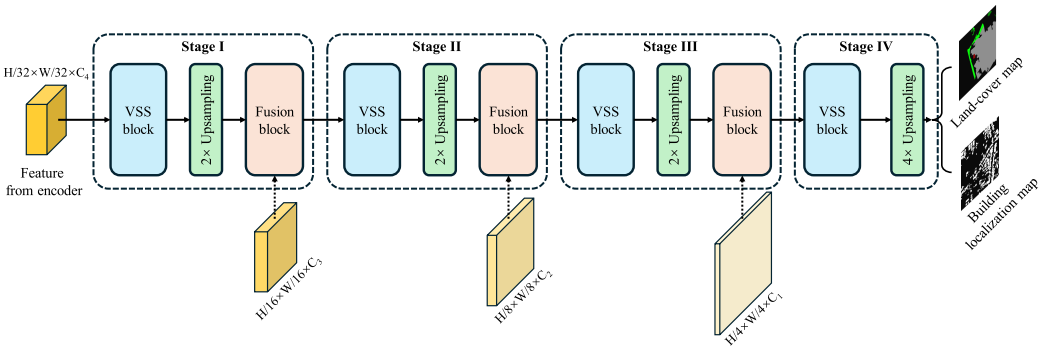


Fig. 6. Specific structure of the semantic decoder in MambaSCD and MambaBDA architectures.

TABLE I
INFORMATION OF THE FIVE BENCHMARK DATASETS USED FOR EXPERIMENTS

Dataset	Study site	Number of image pairs	Size	Resolution	Evaluation task
SYSU	Hong Kong	20,000	256 × 256	0.5m	BCD (category-agnostic)
LEVIR-CD+	Texas of the U.S.	985	1,024 × 1,024	0.5m	BCD (single-category)
WHU-CD	Christchurch, New Zealand	1	32,508 × 15,354	0.3m	BCD (single-category)
SECOND	Hangzhou, Chengdu, and Shanghai	4,662	512 × 512	0.5-3m	SCD
xBD	15 countries	11,034	1,024 × 1,024	<0.8m	BDA

TABLE II
NUMBER OF LAYERS AND FEATURE CHANNELS FOR ENCODERS IN CHANGEMAMBA SERIES

Stage	Tiny	Small	Base
I	$L_1 = 2$ $C_1 = 96$	$L_1 = 2$ $C_1 = 96$	$L_1 = 2$ $C_1 = 128$
II	$L_2 = 2$ $C_2 = 192$	$L_2 = 2$ $C_2 = 192$	$L_2 = 2$ $C_2 = 256$
III	$L_3 = 4$ $C_3 = 384$	$L_3 = 15$ $C_3 = 384$	$L_2 = 15$ $C_3 = 512$
IV	$L_4 = 2$ $C_4 = 768$	$L_4 = 2$ $C_4 = 768$	$L_4 = 2$ $C_4 = 1024$

represent change, facilitating the differentiation of changes from unchanged pixels within the same class.

5) xBD [52]: The xBD dataset, pivotal for advancing BDA in disaster response, integrates pre- and post-disaster satellite imagery with over 850 000 building annotations across 45 362 km^2. The dataset adheres to stringent criteria, featuring high-resolution imagery (better than 0.8 m/pixel), encapsulating multiple levels of damage ("no damage," "minor damage," "major damage," and "destroyed"), and covering a diverse range of disaster types (earthquake, tsunami, storm, wildfire, volcano, and flood).

The information of these five datasets is summarized in Table I.

B. Experimental Setup

1) Implementation Details: All three of our proposed architectures are implemented in Pytorch. Depending on the size and depth of the encoder network, our proposed MambaBCD, MambaSCD, and MambaBDA architectures are available in Tiny, Small, and Base versions [39]. As shown in Table II, the difference between them is the number of VSS blocks inside each stage and the number of channels in the feature map. For datasets other than the SYSU dataset, the multi-temporal image pairs and associated labels are cropped to 256 × 256 pixels for input to the network, and then, the data with the original size are inferred using the trained networks on the test set. During training, we optimize the network using the AdamW optimizer [67] with a learning rate of $1e^{-4}$ and a weight decay of $5e^{-3}$. The batch size is set to 16. Except for the SYSU dataset, for which we set the number of training iterations to 20 000, we set the number of training iterations to 50 000 on the remaining four datasets. Random rotation, left-right, and top-bottom flip are used as training data augmentation methods. Our source code is open sourced for community reproduction and subsequent research in https://github.com/ChenHongruixuan/MambaCD.

2) Evaluation Metrics: For evaluating model performance across different CD subtasks, we employ a set of metrics tailored to the specific requirements of BCD, SCD, and BDA. In evaluating the BCD performance of the model, we use the six commonly used metrics. They are recall rate (Rec), precision rate (Pre), overall accuracy (OA), $F1$ score ($F1$), intersection over union (IoU), and Kappa coefficient (KC). The higher the better for all six indicators. The specific formula of these metrics can be referred to [14], [25], and [45].

For SCD, we follow the existing setup in the literature [56] to use the four following metrics. They are OA, $F1$, mean IoU (mIoU), and separated KC (SeK). The higher the better for all four indicators. Their definitions can be referred to [48] and [56].

For BDA, we follow the protocol of the xView2 Challenge [52] to split BDA into two subtasks, i.e., building localization and damage classification tasks. $F1$ is used to evaluate these two subtasks. First, $F1$ representing the model's

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TABLE III

ACCURACY ASSESSMENT FOR DIFFERENT BCD MODELS ON THE SYSU DATASET. IN THIS AND OTHER TABLES, $\mathcal{C}$ INDICATES CNN-BASED MODELS, $\mathcal{T}$ INDICATES TRANSFORMER-BASED MODELS, AND $\mathcal{M}$ INDICATES MAMBA-BASED MODELS. IN THIS AND OTHER TABLES, WE HIGHLIGHT THE HIGHEST VALUES IN **RED**, AND THE SECOND HIGHEST RESULTS IN **BLUE**

Type	Method	Rec	Pre	OA	F1	IoU	KC
$\mathcal{C}$	FC-EF [22]	75.17	76.47	88.69	75.81	61.04	68.43
	FC-Siam-Diff [22]	75.30	76.28	88.65	75.79	61.01	68.38
	FC-Siam-Conc [22]	76.75	73.67	88.05	75.18	60.23	67.32
	SiamCRNN-18 [17]	76.83	84.80	91.29	80.62	67.54	75.02
	SiamCRNN-34 [17]	76.85	85.13	91.37	80.78	67.75	75.23
	SiamCRNN-50 [17]	78.40	83.41	91.23	80.83	67.82	75.15
	SiamCRNN-101 [17]	**80.48**	80.40	90.77	80.44	67.28	74.40
	SNUNet [41]	72.21	74.09	87.49	73.14	57.66	64.99
	DSIFN [23]	**82.02**	75.83	89.59	78.80	65.02	71.92
	HANet [42]	76.14	78.71	89.52	77.41	63.14	70.59
	CGNet [43]	74.37	**86.37**	91.19	79.92	66.55	74.31
	SEIFNet [68]	78.29	78.61	89.86	78.45	64.54	71.82
$\mathcal{T}$	ChangeFormerV1 [30]	75.82	79.65	89.73	77.69	63.52	71.02
	ChangeFormerV2 [30]	75.62	78.14	89.26	76.86	62.42	69.87
	ChangeFormerV3 [30]	75.24	79.46	89.57	77.29	62.99	70.53
	ChangeFormerV4 [30]	77.90	79.74	90.12	78.81	65.03	72.37
	ChangeFormerV5 [30]	74.00	77.26	88.73	75.60	60.77	68.28
	ChangeFormerV6 [30]	72.38	81.70	89.67	76.76	62.29	70.15
	BIT-18 [28]	76.42	84.85	91.22	80.41	67.24	74.78
	BIT-34 [28]	74.63	82.40	90.26	78.32	64.37	72.06
	BIT-50 [28]	77.90	81.42	90.60	79.62	66.14	73.51
	BIT-101 [28]	75.58	83.64	90.76	79.41	65.84	73.47
	TransUNetCD [34]	77.73	82.59	90.88	80.09	66.79	74.18
	SwinSUNet [32]	79.75	83.50	91.51	81.58	68.89	76.06
	CTDFormer [33]	75.53	80.80	90.00	78.08	64.04	71.61
$\mathcal{M}$	MambaBCD-Tiny	79.59	83.06	91.36	81.29	68.48	75.68
	MambaBCD-Small	78.25	**87.99**	**92.35**	**82.83**	**70.70**	**77.94**
	MambaBCD-Base	80.31	86.11	**92.30**	**83.11**	**71.10**	**78.13**

accuracy in the building localization task is denoted by F_1^{loc}. Then, the model's performance across various damage levels is denoted by $F_1^{\text{level}_i}$. The harmonic mean of all $F_1^{\text{level}_i}$ is calculated to represent as the performance of the model on the damage classification task, formulated as follows:

$$F_1^{\text{clf}} = \frac{1}{\frac{1}{\mathbf{C}^{\text{dam}}} \sum_{i=1}^{\mathbf{C}^{\text{dam}}} \frac{1}{F_1^{\text{level}_i}}} \tag{11}$$

where the value of $\mathbf{C}^{\text{dam}}$ in the xBD dataset is 4, from 1 to 4 being "no damage," "minor damage," "major damage," and "destroyed," respectively. Finally, the overall performance of the model in building localization and damage classification tasks is calculated as $F_1^{\text{overall}} = 0.3\ F_1^{\text{loc}} + 0.7 F_1^{\text{clf}}$.

C. Comparison Methods

We select a number of representative methods based on CNN and Transformer architectures to compare across BCD, SCD, and BDA tasks.

For BCD, our comparison methods include a suite of CNN-based methods: FC-EF [22], FC-Siam-Diff [22], FC-Siam-Conc [22], SiamCRNN [17] (utilizing ResNet architectures ranging from ResNet-18 to ResNet-101 [69]), SNUNet [41], DSIFN [23], HANet [42], CGNet [43], and SEIFNet [68]. For the Transformer-based methods, we compare against ChangeFormer (versions 1–6) [30], BIT (with ResNet-18 to ResNet-101 as the backbone for feature extraction) [28], TransUNetCD [34], SwinSUNet [32], and CTDFormer [33].

For the SCD arena, CNN-based competitors encompass HRSCD (variants S1–S4) [46], ChangeMask [24], SSCD-1 [48], Bi-SRNet [48], and TED [56]. For Transformer-based methods, we evaluate against SMNet [55] and ScanNet [56].

Finally, for the BDA task, we selected Siamese-U-Net [52], MTF [53], and ChangeOS [3] (employing ResNet-18 to ResNet-101) as CNN-based benchmarks. DamFormer [31] stands as the solitary Transformer-based comparison method.

For the above methods, if the method does not have the open source code available, we directly use the accuracy reported in the original paper. Otherwise, we train and test the method on the corresponding dataset based on the hyperparameters recommended in the original paper, using the loss function and data augmentation methods consistent with our method. In addition, in the experimental tables, we will use $\mathcal{C}$ to denote CNN-based models, $\mathcal{T}$ to denote Transformer-based models, and $\mathcal{M}$ to denote Mamba-based models.

D. Detection Results and Benchmark Comparison in Three Tasks

1) BCD: Tables III–V list the performance in BCD of the proposed MambaBCD architecture and comparison methods

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TABLE IV
ACCURACY ASSESSMENT FOR DIFFERENT BINARY CD MODELS ON THE LEVIR-CD+ DATASET

Type	Method	Rec	Pre	OA	F1	IoU	KC
$\mathcal{C}$	FC-EF [22]	71.77	69.12	97.54	70.42	54.34	69.14
	FC-Siam-Diff [22]	74.02	81.49	98.26	77.57	63.36	76.67
	FC-Siam-Conc [22]	78.49	78.39	98.24	78.44	64.53	77.52
	SiamCRNN-18 [17]	84.25	81.22	98.56	82.71	70.52	81.96
	SiamCRNN-34 [17]	83.88	82.28	98.61	83.08	71.05	82.35
	SiamCRNN-50 [17]	81.61	85.39	98.68	83.46	71.61	82.77
	SiamCRNN-101 [17]	80.96	85.56	98.67	83.20	71.23	82.50
	DSIFN [23]	84.36	83.78	98.70	84.07	72.52	83.39
	SNUNet [41]	78.73	71.07	97.83	74.70	59.62	73.57
	HANet [42]	75.53	79.70	98.22	77.56	63.34	76.63
	CGNet [43]	86.02	81.46	98.63	83.68	71.94	82.97
	SEIFNet [68]	81.86	84.83	98.66	83.32	71.41	82.63
$\mathcal{T}$	ChangeFormerV1 [30]	77.00	82.18	98.38	79.51	65.98	78.66
	ChangeFormerV2 [30]	81.32	79.10	98.36	80.20	66.94	79.35
	ChangeFormerV3 [30]	79.97	81.34	98.44	80.65	67.58	79.84
	ChangeFormerV4 [30]	76.68	75.07	98.01	75.87	61.12	74.83
	ChangeFormerV5 [30]	77.96	78.50	98.23	78.23	64.24	77.31
	ChangeFormerV6 [30]	78.57	67.66	97.60	72.71	57.12	71.46
	BIT-18 [28]	80.86	83.76	98.58	82.28	69.90	81.54
	BIT-34 [28]	80.96	85.87	98.68	83.34	71.44	82.66
	BIT-50 [28]	81.84	85.02	98.67	83.40	71.53	82.71
	BIT-101 [28]	81.20	83.91	98.60	82.53	70.26	81.80
	TransUNetCD [34]	84.18	83.08	98.66	83.63	71.86	82.93
	SwinSUNet [32]	85.85	85.34	98.92	85.60	74.82	84.98
	CTDFormer [33]	80.03	80.58	98.40	80.30	67.09	79.47
$\mathcal{M}$	MambaBCD-Tiny	**87.26**	88.82	**99.03**	**88.04**	**78.63**	**87.53**
	MambaBCD-Small	86.49	**89.17**	99.02	87.81	78.27	87.30
	MambaBCD-Base	**87.57**	**89.24**	**99.06**	**88.39**	**79.20**	**87.91**

on the three benchmark datasets. It can be seen that our approaches significantly outperform both the CNN-based approaches and the Transformer-based architectures. The proposed MambaBCD-Base and MambaBCD-Small architectures achieve the highest and second highest OA, $F1$, IoU, and KC for both category-agnostic CD (SYSU dataset) and single-category CD tasks (LEVIR-CD+ and WHU-CD datasets), fully demonstrating the potential of the Mamba architecture for the BCD task. Fig. 7 then shows some binary change maps predicted by our three architectures on the test sets of the three datasets. It can be seen that the proposed methods accurately detect changes with varying types, scales, and numbers contained in these image pairs, very close to the change reference maps.

Table VI lists the network parameters and GFLOPs[2] of these BCD architectures. Benefiting from the S6 model, the MambaBCD architecture can have only linear computational complexity while modeling global context information. It can be seen that even the largest model MambaBCD-Base has GFLOPs of 179.32, which is lower than many CNN and Transformer architectures, such as DSIFN, CFNet, and ChangeFormerV4, comparable with some CNN SOTA methods such as SNUNet and SiamCRNN-50. However, the accuracy advantage is obvious. In particular, the MambaBCD-Tiny architecture has a clear accuracy advantage over other architectures with a similar number of parameters and computational consumption. For example, compared with BIT-18, MambaBCD-Tiny has 56.9% lower GFLOPs but 0.88%, 1.33%, and 3.00% higher $F1$ scores on the three datasets.

2) SCD: Table VII lists the comparison of the SCD performance of the proposed MamabaSCD architectures and representative SCD approaches. Compared with the BCD task, the SCD task requires the change detector capable of effectively modeling not only the spatiotemporal relationships of multitemporal images but also learning representative and discriminative representations for accurate land-cover mapping. From Table VII, we can see that the MambaSCD architecture can satisfy both requirements well. The MambaSCD-Small architecture surpasses the Transformer-based SOTA approach, ScanNet, on all four metrics of the SCD task. Note that our methods have achieved the current accuracy without employing advanced loss functions specifically proposed for the SCD like the ones in ScanNet [56].

Based on the SCD results, the detailed "from–to" semantic change information can be obtained, which is presented in Fig. 8. First, we can find that the "from–to" transition matrices obtained by our methods are very close to the ground truth, demonstrating the effectiveness of our MambaSCD architecture again. Then, based on the contents of the matrices, we can find that the three semantic changes with the highest

[2]https://github.com/Lyken17/pytorch-OpCounter

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TABLE V
ACCURACY ASSESSMENT FOR DIFFERENT BINARY CD MODELS ON THE WHU-CD DATASET

Type	Method	Rec	Pre	OA	F1	IoU	KC
$\mathcal{C}$	FC-EF [22]	86.33	83.50	98.87	84.89	73.74	84.30
	FC-Siam-Diff [22]	84.69	90.86	99.13	87.67	78.04	87.22
	FC-Siam-Conc [22]	87.72	84.02	98.94	85.83	75.18	85.28
	SiamCRNN-18 [17]	90.48	91.56	99.34	91.02	83.51	90.68
	SiamCRNN-34 [17]	89.10	93.88	99.39	91.42	84.20	91.11
	SiamCRNN-50 [17]	91.45	86.70	99.30	90.57	82.76	90.20
	SiamCRNN-101 [17]	90.45	87.79	99.19	89.10	80.34	88.68
	DSIFN [23]	83.45	**97.46**	99.31	89.91	81.67	89.56
	SNUNet [41]	87.36	88.04	99.10	87.70	78.09	87.23
	HANet [42]	88.30	88.01	99.16	88.16	78.82	87.72
	CGNet [43]	90.79	94.47	99.48	92.59	86.21	92.33
	SEIFNet [68]	90.66	91.93	99.36	91.29	83.98	90.96
$\mathcal{T}$	ChangeFormerV1 [30]	84.30	90.80	99.11	87.43	77.67	86.97
	ChangeFormerV2 [30]	83.41	88.77	99.00	86.00	75.45	85.49
	ChangeFormerV3 [30]	85.55	88.25	99.05	86.88	76.80	86.39
	ChangeFormerV4 [30]	84.85	90.09	99.10	87.39	77.61	86.93
	ChangeFormerV5 [30]	84.87	90.20	99.11	87.45	77.70	86.99
	ChangeFormerV6 [30]	81.90	85.49	98.83	83.66	71.91	83.05
	BIT-18 [28]	90.36	90.30	99.29	90.33	82.37	89.96
	BIT-34 [28]	90.10	89.14	99.23	89.62	81.19	89.22
	BIT-50 [28]	90.33	89.70	99.26	90.01	81.84	89.63
	BIT-101 [28]	90.24	89.83	99.27	90.04	81.88	89.66
	TransUNetCD [34]	90.50	85.48	99.09	87.79	78.44	87.44
	SwinSUNet [32]	92.03	94.08	99.50	93.04	87.00	92.78
	CTDFormer [33]	85.37	92.23	99.20	88.67	79.65	88.26
$\mathcal{M}$	MambaBCD-Tiny	91.94	94.76	99.52	93.33	87.49	93.08
	MambaBCD-Small	**92.29**	95.90	**99.57**	**94.06**	**88.79**	**93.84**
	MambaBCD-Base	**92.23**	**96.18**	**99.58**	**94.19**	**89.02**	**93.98**

percentage between the two temporal phases are "nonvegetated ground surface → building", "low vegetation → building," and "low vegetation → nonvegetated ground surface." These results indicate that the areas covered by the SECOND dataset were in the process of urbanization during this period.

Table VIII further lists the number of parameters and GFLOPs of these methods. Compared with the Transformer-based ScanNet method, the GFLOPs of our three architectures are lower. Compared with SMNet, another Transformer-based method, the proposed MamabaSCD-Tiny can achieve higher SeK values with a lower number of parameters and GFLOPs. These results demonstrate the effectiveness of the proposed methods and the potential of the Mamba architecture on the SCD task.

Fig. 9 shows some semantic change maps on the test set of the SECOND dataset. Due to land-cover category combinations, SCD often involves a considerable number of categories. For ease of visualization, we follow the visualization way in [20], [24], and [70], that is, only plotting the land-cover categories of the changed regions in both pre- and post-event images. In this way, the "from–to" semantic change information can be effectively reflected. From Fig. 9, we can observe that the MambaSCD architecture is not only capable of distinguishing changed features at different scales but also accurately identifying the specific semantic categories of these changes.

3) BDA: Table IX reports the accuracy of the MambaBDA architecture and some Transformer and CNN-based architectures on the xBD dataset. The performance of the MambaBDA architecture significantly outperforms these representative BDA approaches. Compared with the current Transformer-based SOTA method, DamFormer [31], our three MambaBDA architectures have a 2.64%, 4.12%, and 4.39% higher F_1^{overall}, respectively. Moreover, it can be noted that the accuracy improvement compared with DamFormer mainly comes from the more difficult damage classification task rather than the building localization task, which demonstrates that the proposed MambaBDA architecture can adequately learn the spatiotemporal relationships between the multitemporal images so as to effectively distinguish different building damage levels.

In order to further show the misclassification between different damage levels in more detail, we show their confusion matrices in Fig. 10. As can be seen from the confusion matrices, the categories "no damage" and "destroyed" are classified with high accuracy and are not easily misclassified into the remaining two categories. On the contrary, the categories of "minor damage" and "major damage" are easily misclassified as other categories, especially the "minor damage" category. A large portion of the "minor damage" category is misclassified as "no damage" and "major damage." There are mainly the following reasons. First, the optical images in the xBD

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TABLE VI
COMPARISON OF DIFFERENT BINARY CD MODELS IN COMPUTATIONAL COST

Type	Method	Params (M)	GFLOPs
$\mathcal{C}$	FC-EF [22]	1.35	14.13
	FC-Siam-Diff [22]	1.35	18.66
	FC-Siam-Conc [22]	1.54	21.07
	SiamCRNN-18 [17]	18.85	86.64
	SiamCRNN-34 [17]	28.96	113.28
	SiamCRNN-50 [17]	44.45	185.30
	SiamCRNN-101 [17]	63.44	224.30
	DSIFN [23]	35.73	329.03
	SNUNet [41]	10.21	176.36
	HANet [42]	2.61	70.68
	CGNet [43]	33.68	329.58
	SEIFNet [68]	39.08	167.75
$\mathcal{T}$	ChangeFormerV1 [30]	29.84	46.62
	ChangeFormerV2 [30]	24.30	44.54
	ChangeFormerV3 [30]	24.30	33.68
	ChangeFormerV4 [30]	33.61	852.53
	ChangeFormerV5 [30]	55.27	841.08
	ChangeFormerV6 [30]	41.03	811.15
	BIT-18 [28]	11.50	106.14
	BIT-34 [28]	21.61	190.83
	BIT-50 [28]	24.28	224.61
	BIT-101 [28]	43.27	380.62
	TransUNetCD [34]	28.37	244.54
	SwinSUNet [32]	39.28	43.50
	CTDFormer [33]	3.85	303.77
$\mathcal{M}$	MambaBCD-Tiny	17.13	45.74
	MambaBCD-Small	49.94	114.82
	MambaBCD-Base	84.70	179.32

TABLE VII
ACCURACY ASSESSMENT FOR DIFFERENT SCD MODELS ON THE SECOND DATASET. NOTE THAT UNLIKE [24], [48], AND [56] AND OTHERS THAT DIVIDE THE TRAINING SET FOR TESTING, WE TRAIN ON THE TRAINING SET AND THEN TEST ON THE TEST SET

Type	Method	OA	F1	mIoU	SeK_{37}
$\mathcal{C}$	HRSCD-S1 [46]	62.56	33.04	47.57	6.62
	HRSCD-S2 [46]	82.93	39.79	63.80	8.46
	HRSCD-S3 [46]	81.17	41.78	64.96	10.86
	HRSCD-S4 [46]	83.88	47.68	70.58	16.81
	ChangeMask [24]	85.53	50.54	71.39	18.36
	SSCD-1 [48]	85.62	52.80	72.14	20.15
	BiSRNet [48]	85.80	53.59	72.69	21.18
	TED [56]	85.50	52.61	72.19	20.29
$\mathcal{T}$	SMNet [55]	84.29	50.48	71.62	18.98
	ScanNet [56]	86.13	55.21	72.48	21.57
$\mathcal{M}$	MamabaSCD-Tiny	**90.44**	54.51	73.33	22.08
	MamabaSCD-Small	86.48	**55.72**	**73.41**	**22.83**
	MamabaSCD-Base	**90.36**	**55.32**	**73.68**	**22.92**

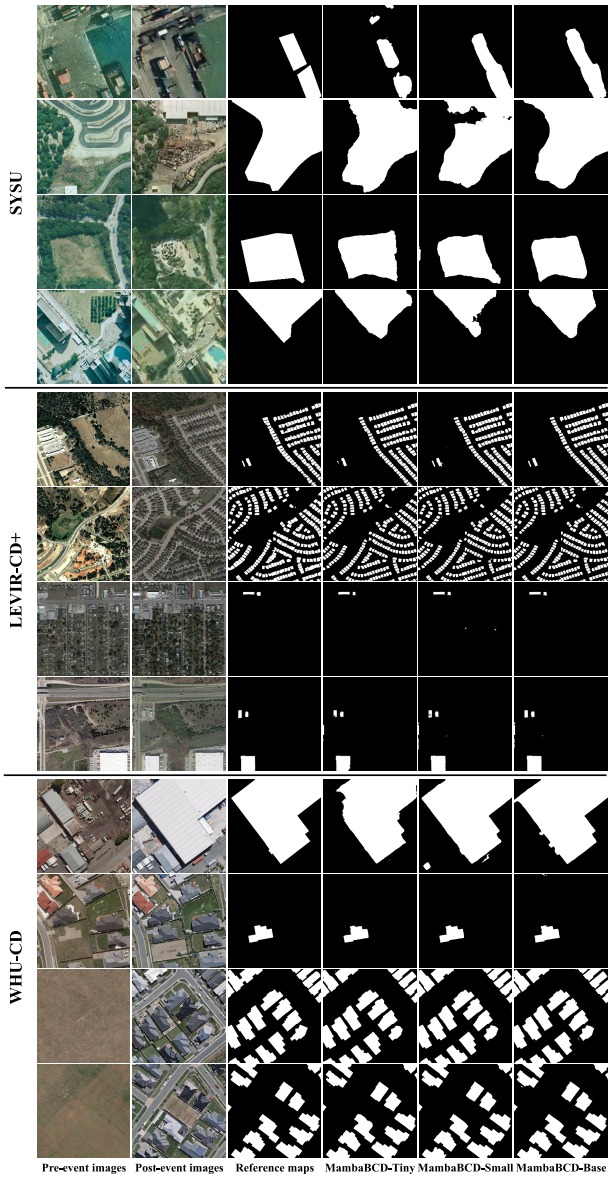


Fig. 7. Some binary change maps obtained by our methods on the test set of the three BCD dataset.

TABLE VIII
COMPARISON OF DIFFERENT SCD MODELS IN COMPUTATIONAL COST

Type	Method	Params	GFLOPs
$\mathcal{C}$	HRSCD-S1 [46]	3.36	8.02
	HRSCD-S2 [46]	6.39	14.29
	HRSCD-S3 [46]	12.77	42.67
	HRSCD-S4 [46]	13.71	43.69
	ChangeMask [24]	2.97	37.16
	SSCD-1 [48]	23.31	189.57
	Bi-SRNet [48]	23.39	189.91
	TED [56]	24.19	204.29
$\mathcal{T}$	SMNet [55]	42.16	75.79
	ScanNet [56]	27.90	264.95
$\mathcal{M}$	MamabaSCD-Tiny	21.51	73.42
	MamabaSCD-Small	54.28	146.70
	MamabaSCD-Base	89.99	211.55

dataset were acquired with a relatively small viewing angle of the sensor. In many cases, "minor damage" is nuance damage to the structure of the building. Little relevant information can be obtained from the nearly orthorectified optical image for such cases. Furthermore, there is a semantic ambiguity in the definition of the categories of minor damage and major damage. We argue that it is difficult to address this problem by starting from the model architecture perspective. It can be

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TABLE IX
ACCURACY ASSESSMENT FOR DIFFERENT BDA MODELS ON THE xBD DATASET. THE SUFFIX PPS MEANS POSTPROCESSING

Type	Method	F_1^{loc}	F_1^{clf}	$F_1^{overall}$	Damage F_1 per class			
					No damage	Minor damage	Major damage	Destroyed
$\mathcal{C}$	Siamese-UNet [52]	85.92	65.58	71.68	86.74	50.02	64.43	71.68
	MTF [53]	83.60	70.02	74.10	90.60	49.30	72.20	83.70
	ChangeOS-18 [3]	84.62	69.87	74.30	88.61	52.10	70.36	79.65
	ChangeOS-34 [3]	85.16	70.28	74.74	88.63	52.38	71.16	80.08
	ChangeOS-50 [3]	85.41	70.88	75.24	88.98	53.33	71.24	80.60
	ChangeOS-101 [3]	85.69	71.14	75.50	89.11	53.11	72.44	80.79
	ChangeOS-18-PPS [3]	84.62	73.89	77.11	92.38	57.41	72.54	82.62
	ChangeOS-34-PPS [3]	85.16	74.25	77.52	92.19	58.07	72.84	82.79
	ChangeOS-50-PPS [3]	85.41	75.64	78.57	92.66	60.14	74.18	83.45
	ChangeOS-101-PPS [3]	85.69	75.44	78.52	92.81	59.38	74.65	83.29
$\mathcal{T}$	DamFormer [31]	86.86	72.81	77.02	89.86	56.78	72.56	80.51
$\mathcal{M}$	MambaBDA-Tiny	**87.20**	78.30	80.97	95.84	60.96	**77.43**	88.23
	MambaBDA-Small	86.61	**78.80**	**81.14**	**95.99**	**62.82**	76.26	**88.37**
	MambaBDA-Base	**87.38**	**78.84**	**81.41**	**95.94**	**62.74**	**76.46**	**88.58**

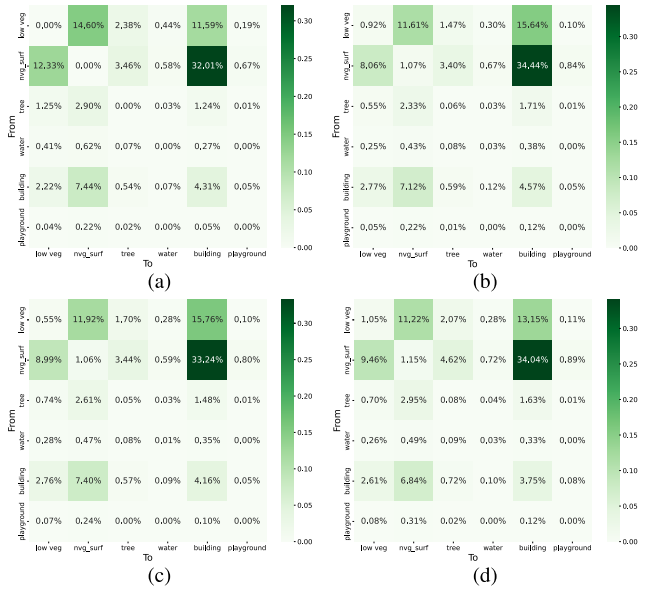


Fig. 8. Semantic change matrices providing "from–to" transition information derived from the SCD results. (a) Reference. (b) MambaSCD-Tiny. (c) MambaSCD-Small. (d) MambaSCD-Base.

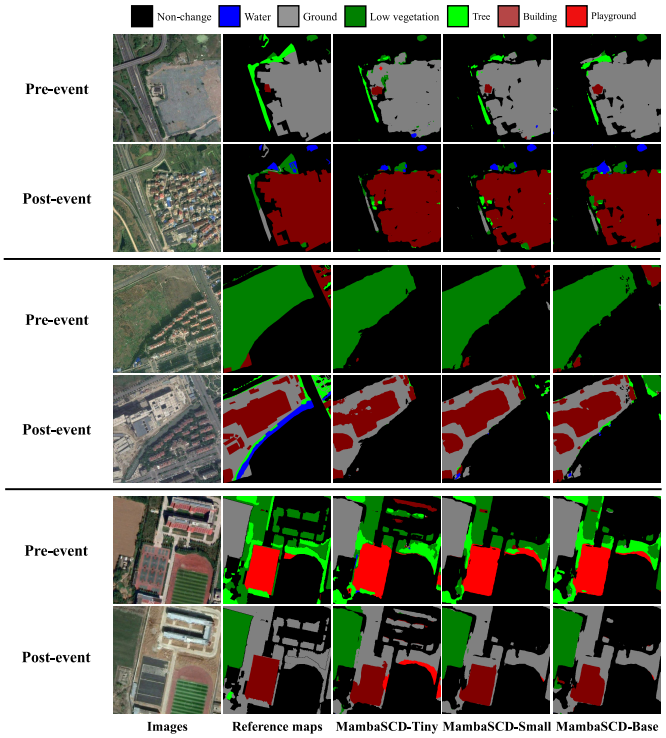


Fig. 9. Some semantic change maps obtained by the proposed MamabaSCD architectures on the test set of the SECOND dataset.

considered to introduce auxiliary information, such as height information or multiview imagery [21], [71] to help the model make a more accurate classification.

Table X lists the number of parameters and GFLOPs of these BDA approaches. For MambaBDA-Small, our architecture yields more accurate results on BDA compared with Siamese-UNet, ChangeOS-101, and MTF, with an improvement in F_1^{overall} from 5.64 % to 9.46 % for a similar number of parameters and lower GFLOPs.

Fig. 11 shows the visualization results of the building damage caused by different disaster events obtained by MambaBDA on the xBD dataset. MambaBDA can accurately localize and then differentiate the damage levels of buildings despite the varying type of building and the disaster event, implying the potential of the MambaBDA architecture for practical disaster response applications.

E. Different Spatiotemporal Modeling Methods

Modeling spatiotemporal relationships is significant for CD tasks [17], [24], [47]. To demonstrate the effectiveness of our proposed modeling mechanisms, we compare them with several commonly used spatial–temporal modeling methods in CD. They are concatenation operations with feature pyramid network (FPN) [24] (baseline), temporal-symmetric transformation based on 3-D convolutional layers (TST) [24],

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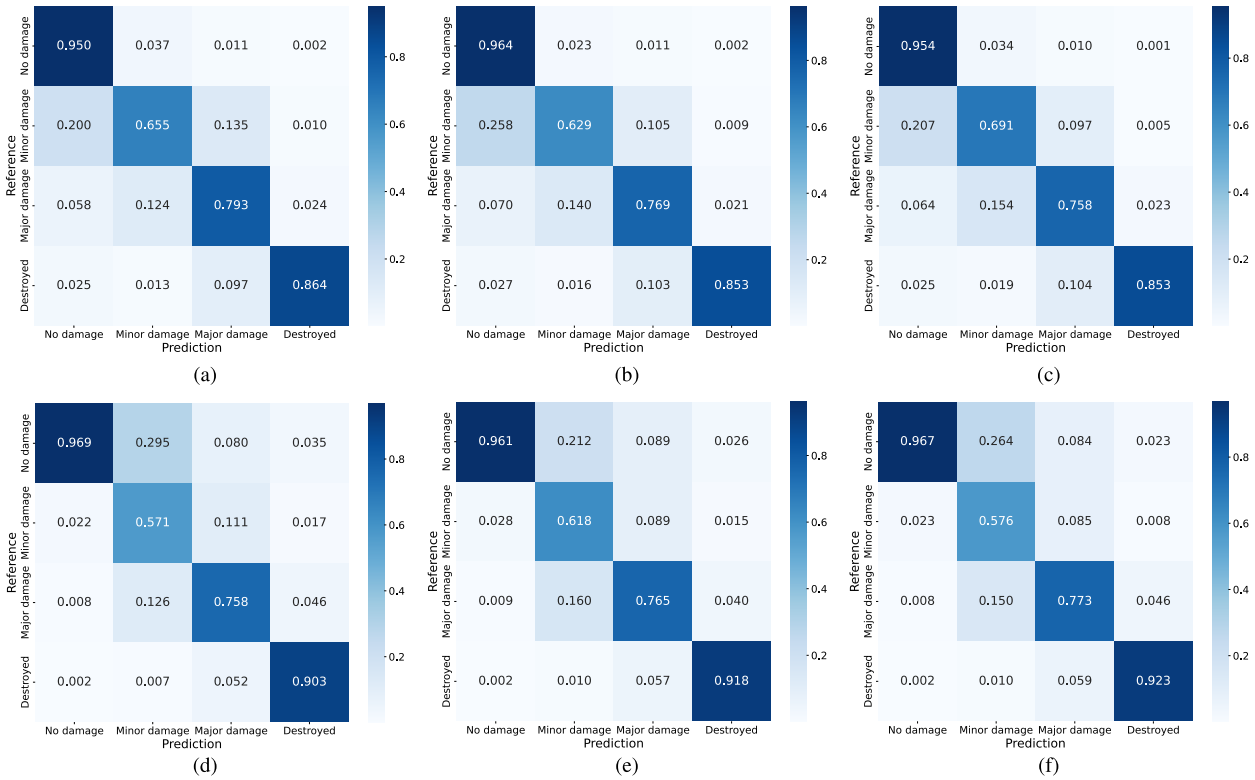


Fig. 10. Confusion matrices of MambaBDA-Tiny, MambaBDA-Small, and MambaBDA-Base. (a)–(c) Matrices normalized by rows. (d)–(f) Matrices normalized by columns.

TABLE X
COMPARISON OF DIFFERENT BDA MODELS IN COMPUTATIONAL COST

Type	Method	Params	GFLOPs
$\mathcal{C}$	Siamese-UNet [52]	51.46	138.88
	MTF [53]	44.40	268.76
	ChangeOS-18 [3]	25.25	91.55
	ChangeOS-34 [3]	35.36	110.89
	ChangeOS-50 [3]	39.05	118.41
	ChangeOS-101 [3]	58.05	157.26
$\mathcal{T}$	DamFormer [31]	32.52	169.34
$\mathcal{M}$	MamabaBDA-Tiny	19.74	59.57
	MamabaBDA-Small	52.11	130.80
	MamabaBDA-Base	87.76	195.43

RNN-based method [17], and Tranformer-based method [28]. Table XI lists the performance of these modeling methods on the three CD subtasks. On the simpler BCD tasks, Transformer does not yield significant advantages in performance compared with RNN. However, on the more difficult SCD and BDA tasks, the advantages of the Transformer architecture are illustrated. Compared with these current representative spatiotemporal relationship modeling approaches, our proposed modeling approach based on the Mamba architecture, which inherits the advantages of RNN and Transformer, shows superior performance on the three CD tasks of BCD, SCD, and BDA. Compared with the Transformer-based spatiotemporal modeling approach, our method improves the $F1$ value by 0.77% on BCD, SeK by 0.52% on the SCD task and F_1^{overall} by 1.10% on the BDA task.

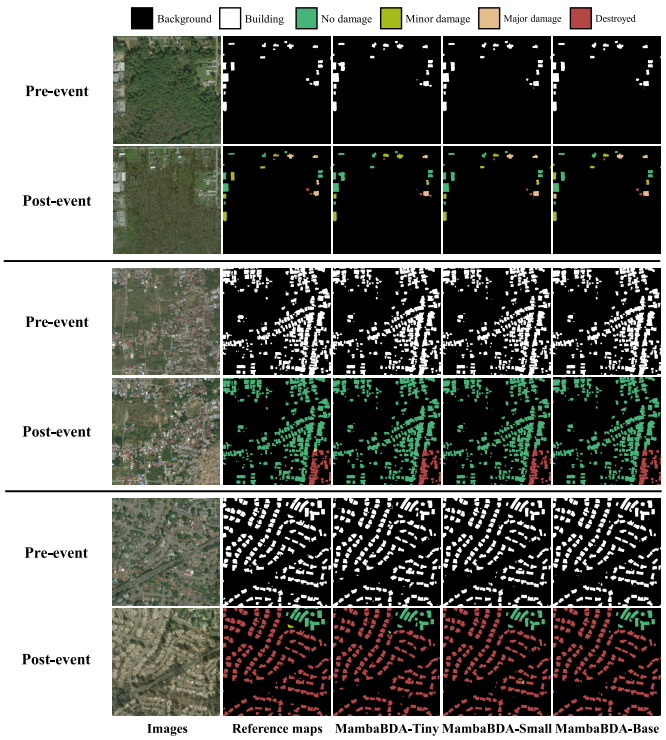


Fig. 11. Some BDA results obtained by our methods on the test set of the xBD dataset. From top to bottom, the disaster events are typhoons, tsunamis, and wildfires.

In Table XII, we list the detection performance obtained using one of the three modeling mechanisms alone. First, using any of the modeling mechanisms can significantly improve

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TABLE XI
COMPARISON OF DIFFERENT APPROACHES FOR MODELING SPATIOTEMPORAL RELATIONSHIPS ON THREE TASKS

Method	SYSU				SECOND				xBD		
	OA	F1	IoU	KC	OA	F1	mIoU	SeK	F_1^{loc}	F_1^{clf}	$F_1^{overall}$
Baseline	91.71	81.11	68.22	75.84	86.11	54.20	72.42	21.15	84.85	74.74	77.77
STT [24]	92.09	**82.22**	**69.80**	77.16	86.35	54.70	72.86	21.83	**86.74**	75.92	79.17
RNN [17]	**92.97**	82.02	69.52	76.94	86.46	54.76	72.89	21.90	85.74	76.48	79.26
Transformer [28]	92.14	82.17	69.74	**77.17**	**86.66**	**55.50**	**73.01**	**22.31**	86.09	**77.45**	**80.04**
Ours	**92.35**	**82.83**	**70.70**	**77.94**	**86.48**	**55.72**	**73.41**	**22.83**	**86.61**	**78.80**	**81.14**

TABLE XII
ABLATION STUDY OF THE PROPOSED SPATIOTEMPORAL MODELING MECHANISMS ON THE SYSU DATASET

Modelling Mechanism			OA	F1	IoU	KC
Sequential	Cross	Parallel				
			91.71	81.11	68.22	75.84
✓			92.16	82.20	69.78	77.20
	✓		**92.20**	**82.24**	**69.84**	**77.28**
		✓	92.12	82.17	69.73	77.14
✓	✓	✓	**92.35**	**82.83**	**70.70**	**77.94**

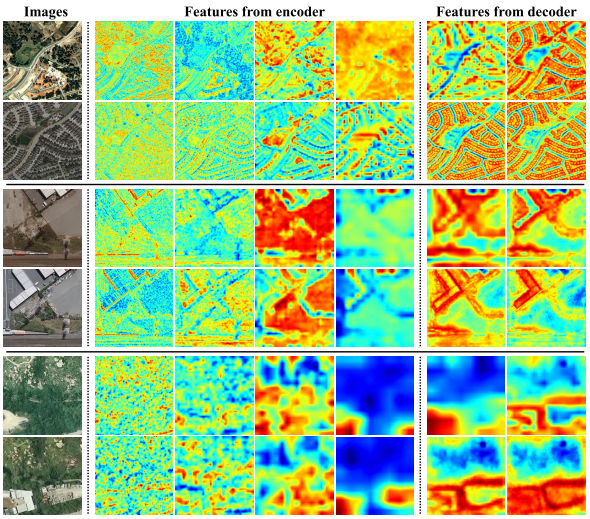


Fig. 12. Visualization of features from different layers of MambaBCD.

detection performance compared with the baseline. Then, it can be seen that the performance gap between the three modeling mechanisms is not significant, with spatiotemporal cross modeling being slightly better than the other 2. Finally, the detection performance of the network can be further improved by using the three modeling mechanisms simultaneously.

We also visualize the features extracted by the MambaBCD architecture in Fig. 12. In the features extracted by the encoder, the features in the region of high response values are not necessarily regions of interest, i.e., changed regions. Whereas in the decoder, after the spatiotemporal relationships are modeled by the proposed spatiotemporal modeling mechanisms, changed regions gradually exhibit high response values.

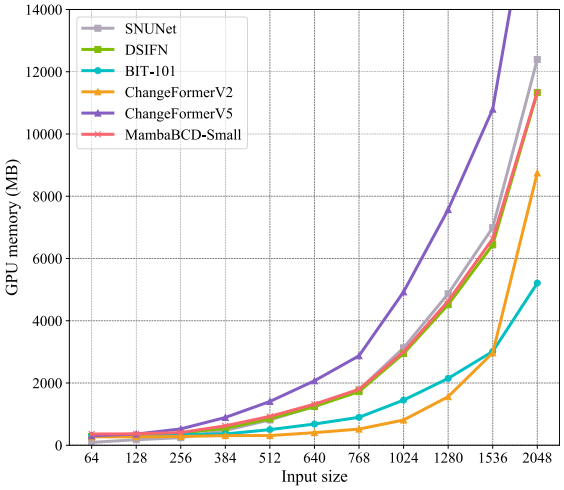


Fig. 13. GPU memory footprint of different models with varying input sizes.

F. Comparison With Other Backbone Networks

We also simply compare the Mamba with some representative backbone networks in CD tasks on the SYSU dataset in Table XIII. It can be seen that Mamba outperforms the two Transformer backbone networks, Swin-Transformer and MixFormer. This is because, in order to reduce the computational overhead of self-attention operation, these two Transformer backbone networks either adopt a local attention mechanism (Swin-Transformer) [35] or a strided convolutional layer (MixFormer) [36] to reduce the size of the feature maps, which will negatively affect the learning of global contextual information. The adopted VMamba architecture, on the other hand, can reduce its computational consumption to a linear relationship with the number of tokens, thus eliminating the need for these methods, which cause information loss, and allowing for the full learning of global context information. As for the remote sensing CD task, due to the multiscale phenomenon of land-cover features and the differences in spatial resolution of different sensors, the sizes of changed objects often vary greatly. Therefore, the features extracted by Mamba through fully learning the global contextual information are particularly important for the subsequent detection of these changed objects with different sizes.

G. GPU Memory Footprint

In Section IV-D, we list the number of parameters and GFLOPs of different models. In this section, we further

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TABLE XIII
COMPARISON OF DIFFERENT COMMONLY USED BACKBONES IN BCD ON THE SYSU DATASET

Encoder	OA	F1	IoU	KC	Param (M)	GFLOPs
ResNet-101 [69]	91.10	81.10	68.22	75.28	46.90	131.36
EfficientNet-B5 [72]	91.60	**82.26**	**69.86**	76.84	31.63	76.88
MixFormer-v3 [36]	87.70	75.00	60.00	66.87	47.03	82.98
Swin-Small [35]	**92.09**	82.22	69.80	**77.16**	37.75	97.99
VMamba-Small [39]	**92.35**	**82.83**	**70.70**	**77.94**	49.94	114.82

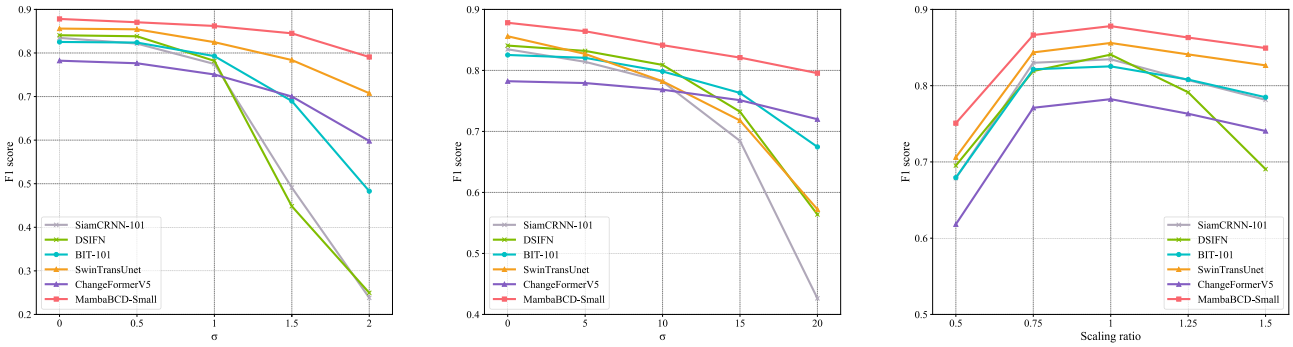


Fig. 14. Robustness of different models against degraded data on the LEVIR-CD+ dataset. (a) Gaussian blur. (b) Gaussian noise. (c) Scaling.

compare the GPU memory footprint of some models. Fig. 13 shows the comparison of the GPU memory footprint of some CNN-based and Transformer-based methods with our MambaBCD architecture for varying input sizes. We can see that the trend of GPU memory footprint with input size for MambaBCD-Small is similar to some of the advanced CNN architectures (DSIFN and SNUNet). However, our MambaBCD is able to learn the global spatial contextual information as well as adequately model spatiotemporal relationships, thereby exhibiting better detection performance. The GPU memory footprint of MambaBCD-Small is significantly lower compared with ChangeFormerV5, which employs the Transformer architecture as its backbone. Note that ChangeFormerV5 itself has employed strided convolutional layers to reduce the computational overhead of self-attention operations. Also, we can find that the lightweight architecture ChangeFormerV2 has surpassed the Mamba architecture and the two CNN architectures, DSIFN and SNUNet, in terms of the increasing trend of GPU memory footprint when the input size is increased from 1536 to 2048. These results imply that Transformer to be applicable to large-scale scenarios of remote sensing images often comes at the cost of information loss. In comparison, Mamba architecture can avoid this problem.

H. Robustness Against Degraded Input Data

Finally, we test the robustness of our method to degraded input data. In the inference phase, the data input to the model is subjected to different levels of Gaussian blurring, Gaussian noise, and scaling of the data by different proportions, respectively. Fig. 14 shows the degradation of the $F1$ score of different methods with degraded data on the LEVIR-CD+ dataset. First, we can see that the CNN-based methods are very poorly robust to Gaussian blurred data. $F1$ scores of SiamCRNN-101 and DSIFN drop by nearly 60% when increasing the standard deviation of the Gaussian blur from 0 to 2. Whereas the two methods using Transformer as the backbone, SwinTransUNet and ChangeFormerV5 are relatively robust to Gaussian blurred data, with $F1$ scores decreasing by 14.83 % and 18.43 %, respectively. In contrast, the $F1$ score for our MambaBCD architecture drops by only 8.73%.

Fig. 14(b) shows the effect of adding different levels of Gaussian noise to the input data. With the exception of ChangeFormerV5 and our MambaBCD, the remaining methods have significant $F1$ decreases ranging from 40.83% to 15.07%. Our MambaBCD architecture, on the other hand, shows robustness to Gaussian noise, with only an 8.26% drop in the $F1$ score for Gaussian noise with a standard deviation of 20.

Fig. 14(c) shows the comparison of the robustness of models trained at a single scale to test data at different scales. The $F1$ value of MambaBCD decreases by only 2.88% when the scale increases from 1 to 1.5. When the scaling ratio decreases from 1 to 0.75, the $F1$ value of MambaBCD decreases only by 1.18%. When the scaling factor is reduced to 0.5, all methods, including MambaBCD, show a larger decrease in the $F1$ score. However, compared with the decline of 14.99 % of SwinTransUNet and the decline of 16.41 % of ChangeFormerV5, the decline of the MambaBCD decline is smaller, at 12.03 %. All the above results again demonstrate the superiority of our proposed Mamba-based CD architecture.

V. CONCLUSION

In this article, we explore the potential of the emerging Mamba architecture for remote sensing image CD tasks. Three architectures, MambaBCD, MambaSCD, and MambaBDA are

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developed for BCD, SCD, and BDA tasks, respectively. The encoders of all three architectures are cutting-edge VMamba architecture, which can adequately learn the global contextual information of the input image with linear complexity. Then, for the change decoder, in order to adequately learn spatiotemporal features, we propose three ways of modeling spatiotemporal relationships that can adequately exploiting the attributes and advantages of Mamba architecture. Experiments on five benchmark datasets have fully revealed the potential of Mamba for multitemporal remote sensing image data processing tasks. Compared with the CNN-based and Transformer-based approaches, the proposed three architectures can achieve quite competitive performance on their respective tasks. Compared with some commonly used backbone networks, the Mamba architecture can extract features that are more suitable for downstream CD tasks. The combination of our proposed three spatiotemporal relationship modeling mechanisms and the Mamba architecture provides a new perspective on modeling spatiotemporal relationships, showing much better detection results on all three subtasks.

Our future work includes, but is not limited to, developing Mamba architectures more suited to the characteristics of remote sensing data, applying them to multimodal and time-series remote sensing tasks, and exploring the potential of Mamba for remote sensing foundation models.

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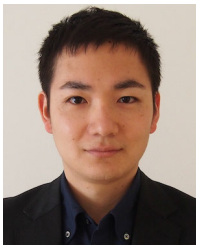
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